

as the cross terms equate to zero. By the Rayleigh quotient bound, $g^\top \mathbf{F} g \leq \lambda_1 \|g\|^2 = \lambda_1 \|Pg\|^2$. On $V_d^\perp$, the largest eigenvalue is λ_{d+1} , thus by the same technique $g^\top \mathbf{F} g \leq \lambda_{d+1} (1 - \|Pg\|^2)$ using $\|g\| = 1$ and orthogonality. Summation yields

$$g^\top \mathbf{F} g \leq \lambda_1 \|Pg\|^2 + \lambda_{d+1} (1 - \|Pg\|^2)$$

with the final inequality following from $(1 - \|Pg\|^2) \leq 1$. $\square$

We now collect the lemmas and prove Theorem 4.1

Proof of Theorem 4.1. Fix a skill S_i and write $\mathbf{F} := \mathbf{F}_i(\theta^*) \succeq 0$. Let $\Delta\theta = \eta g$ where $g \sim \text{Unif}(\mathbb{S}^{n-1})$ and $\eta > 0$ is sufficiently small so that the local quadratic approximation applies. By the second-order expansion of the skill utility (cf. Lemma B.1),

$$\Delta u_i = \frac{1}{2} \Delta\theta^\top \mathbf{F} \Delta\theta + O(\|\Delta\theta\|^3) = \frac{\eta^2}{2} g^\top \mathbf{F} g + O(\eta^3), \quad (2)$$

since $\|g\| = 1$.

We begin with proving the expectation result. By Lemma B.2, for any symmetric matrix $\mathbf{F}$,

$$\mathbb{E}[g^\top \mathbf{F} g] = \frac{1}{n} \text{Tr}(\mathbf{F}). \quad (3)$$

Taking expectations yields

$$\mathbb{E}[\Delta u_i] = \frac{\eta^2}{2n} \text{Tr}(\mathbf{F}_i(\theta^*)) + O(\eta^3), \quad (4)$$

establishing part (1) of the theorem.

We now convert this to a high-probability result. Let P denote the orthogonal projector onto the span of the top d eigenvectors of $\mathbf{F}$, and let $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ be the eigenvalues of $\mathbf{F}$. By Lemma B.3, for any $\varepsilon \in (0, 1)$,

$$\Pr\left[\|Pg\|^2 \leq (1 + \varepsilon) \frac{d}{n}\right] \geq 1 - 2e^{-c\varepsilon^2 d}. \quad (5)$$

Condition on this event. Lemma B.4 then implies that for any unit vector g ,

$$g^\top \mathbf{F} g \leq \lambda_1 \|Pg\|^2 + \lambda_{d+1} (1 - \|Pg\|^2). \quad (6)$$

Substituting the concentration bound yields, with probability at least $1 - 2e^{-c\varepsilon^2 d}$,

$$g^\top \mathbf{F} g \leq \lambda_1 (1 + \varepsilon) \frac{d}{n} + \lambda_{d+1} \left(1 - (1 + \varepsilon) \frac{d}{n}\right) \quad (7)$$

$$\leq \lambda_1 (1 + \varepsilon) \frac{d}{n} + \lambda_{d+1}. \quad (8)$$

Plugging into the expansion for Δu_i gives

$$\Delta u_i \leq \frac{\eta^2}{2} \left(\lambda_{\max}(\mathbf{F}_i(\theta^*)) (1 + \varepsilon) \frac{d}{n} + \lambda_{d+1} \right) + O(\eta^3). \quad (9)$$

In particular, when the curvature of $\mathbf{F}_i(\theta^*)$ is concentrated in its top d eigenspace (i.e., λ_{d+1} is negligible), this implies the stated scaling

$$\Delta u_i = O\left(\frac{d\eta^2}{n} (1 + \varepsilon) \lambda_{\max}(\mathbf{F}_i(\theta^*))\right), \quad (10)$$

with probability at least $1 - 2e^{-c\varepsilon^2 d}$, completing the proof. $\square$

A.3 Proofs from Section 6

Proof of Theorem 6.1. Under Assumption 1, θ^* maximizes u_i , implying the gradient vanishes. The expansion for the loss is therefore:

$$\Delta u_i(\theta) = \frac{1}{2} \Delta \theta^\top \mathbf{F}_i(\theta^*) \Delta \theta - O(\|\Delta \theta\|^3) \quad (11)$$

as before. By definition of this utility and the fact that $\|\mathbf{F}_i^{1/2} \Delta \theta\|^2 \geq \|\mathbf{F}_i^{1/2} P_i(\Delta \theta)\|^2$ since P_i is an orthogonal projector we have the first bound of the theorem. We now need to ensure that the quadratic cost of moving in a sensitive direction is not reduced by moving in an orthogonal direction. Specifically, the following lemma shows that if the subspaces are defined by their eigenvectors, then the interaction between sensitive and non-sensitive components of the update vector is exactly zero.

Lemma A.5. *Let $\mathbf{F} \in \mathbb{R}^{n \times n}$ be symmetric positive semidefinition with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ and orthonormal eigenvectors $v_1, \dots, v_n$. Fix $d \in [n]$ and let $M = \text{span}\{v_1, \dots, v_d\}$. Let P be the orthogonal projector onto M . If $\lambda_d \geq \lambda$ for some $\lambda > 0$, then for all $z \in \mathbb{R}^n$, $z^\top \mathbf{F} z \geq \lambda \|Pz\|^2$.*

Proof. Expand z in the eigenbasis: $z = \sum_{j=1}^n a_j v_j$ where $a_j = \langle z, v_j \rangle$. Then

$$\begin{aligned} z^\top \mathbf{F} z &= \sum_{j=1}^n \lambda_j a_j^2 \\ &\geq \sum_{j=1}^d \lambda_j a_j^2 \\ &\geq \lambda \sum_{j=1}^d a_j^2 \end{aligned}$$

On the other hand, $Pz = \sum_{j=1}^d a_j v_j$, hence $\|Pz\|^2 = \sum_{j=1}^d a_j^2$. Thus, we have the result. $\square$

Combining the above inequality with the expansion of our loss function for all θ in the neighborhood of θ^* , we obtain

$$\Delta u_i \geq \frac{\lambda}{2} \|P_i(\Delta \theta)\|^2 - C \|\Delta \theta\|^3$$

Consequently, if $\|P_i(\Delta \theta)\| \geq \delta$, then the bound becomes

$$\Delta u_i \geq \frac{\lambda}{2} \delta^2 - C \|\Delta \theta\|^3.$$

Moreover for $\|\Delta \theta\| \leq \frac{\lambda \delta^2}{4C}$ we obtain the simplified $\Delta u_i = \Omega(\lambda \delta^2)$. $\square$

Proof of Theorem 6.2. Since g is C^2 on the closed ball $B(\theta^*, r)$, it is locally Lipschitz and bounded there. By standard existence and uniqueness results for ordinary differential equations [], the gradient flow $\dot{\theta}(t) = -g(\theta(t))$, $\theta(0) = \theta^*$ admits a unique solution on $[0, t_0]$ for some $t_0 > 0$, and by choosing $t_0 \leq r / \sup_{B(\theta^*, r)} \|g\|$, the trajectory remains in the ball for all $t \in [0, t_0]$. Moreover, Taylor expanding $\theta(t)$ at $t = 0$ and using the boundedness of the gradient and Hessian on the ball, we obtain:

$$\theta(t) - \theta^* = -tg(\theta^*) + \frac{t^2}{2} \nabla g(\theta^*) g(\theta^*) + O(t^3).$$

We proceed to prove the following Fisher-weighted projector bound stemming from the AIC.

Lemma A.6. Let $\mathbf{F}_i := \mathbf{F}_i(\theta^*)$ and let P_i be the orthogonal projector onto M_i . Assume the second-order expansion above holds and write $R(t)$ for the remainder so that $\|R(t)\| \leq Ct^3$. Then for all $t \in [0, t_0]$,

$$\|\mathbf{F}_i^{1/2} P_i(\theta(t) - \theta^*)\| \geq \frac{t^2}{2} \|\mathbf{F}_i^{1/2} P_i \nabla g(\theta^*) g(\theta^*)\| - t \|\mathbf{F}_i^{1/2} P_i g(\theta^*)\| - C't^3, \quad (12)$$

for some constant $C' > 0$ (depending on $\mathbf{F}_i$ and local smoothness of g). In particular, under the AIC,

$$\|\mathbf{F}_i^{1/2} P_i(\theta(t) - \theta^*)\| \geq \frac{\gamma}{2} t^2 - \sqrt{\lambda_{\max}(\mathbf{F}_i)} \varepsilon t - C't^3.$$

Proof. Apply the linear map $\mathbf{F}_i^{1/2} P_i$ to the second-order expansion:

$$\mathbf{F}_i^{1/2} P_i(\theta(t) - \theta^*) = \underbrace{\frac{t^2}{2} \mathbf{F}_i^{1/2} P_i (\nabla g(\theta^*) g(\theta^*))}_A + \underbrace{\left(-t \mathbf{F}_i^{1/2} P_i g(\theta^*) + \mathbf{F}_i^{1/2} P_i R(t) \right)}_B.$$

Write $\mathbf{F}_i^{1/2} P_i(\theta(t) - \theta^*) = A + B$. By the reverse triangle inequality, $\|A + B\| \geq \|A\| - \|B\|$.

We bound B . By the triangle inequality,

$$\|B\| \leq t \|\mathbf{F}_i^{1/2} P_i g(\theta^*)\| + \|\mathbf{F}_i^{1/2} P_i R(t)\|.$$

Unlike P_i , the map $\mathbf{F}_i^{1/2} P_i$ need not be nonexpansive, so we bound it by its operator norm:

$$\|\mathbf{F}_i^{1/2} P_i R(t)\| \leq \|\mathbf{F}_i^{1/2} P_i\| \|R(t)\| \leq \|\mathbf{F}_i^{1/2}\| \|R(t)\| = \sqrt{\lambda_{\max}(\mathbf{F}_i)} \|R(t)\| \leq \sqrt{\lambda_{\max}(\mathbf{F}_i)} Ct^3.$$

Absorb $\sqrt{\lambda_{\max}(\mathbf{F}_i)}C$ into a constant C' to get $\|B\| \leq t \|\mathbf{F}_i^{1/2} P_i g(\theta^*)\| + C't^3$.

Also, $\|A\| = \frac{t^2}{2} \|\mathbf{F}_i^{1/2} P_i \nabla g(\theta^*) g(\theta^*)\|$. Combining these inequalities yields (12).

Finally, under the AIC, $\|\mathbf{F}_i^{1/2} P_i \nabla g(\theta^*) g(\theta^*)\| \geq \gamma$. Moreover, since $\|P_i g(\theta^*)\| \leq \varepsilon$ (AIC Condition 2) and $\|\mathbf{F}_i^{1/2} P_i g(\theta^*)\| \leq \|\mathbf{F}_i^{1/2}\| \|P_i g(\theta^*)\| = \sqrt{\lambda_{\max}(\mathbf{F}_i)} \varepsilon$, substituting gives the stated bound. $\square$

The lemma together with the regularity conditions yields the theorem. $\square$

Proof of Corollary 6.3. By Theorem 6.2, given the conditions on t , we have

$$\|\mathbf{F}_i^{1/2} P_i(\theta(t) - \theta^*)\| = \Omega(\gamma t^2),$$

up to lower-order terms (in particular, once the quadratic term dominates the linear term). Substituting this into Theorem 6.1 and using the local bound $\|\theta(t) - \theta^*\| = O(t)$ along the gradient flow yields

$$\Delta u_i(\theta(t)) \geq \frac{1}{2} \|\mathbf{F}_i^{1/2} P_i(\theta(t) - \theta^*)\|^2 - O(t^3) = \Omega(\gamma^2 t^4),$$

for all t in the stated regime and some $t_0 > 0$. $\square$

A.4 Connection to Sensitive Subspace in Representation and Low Rankness of Weight Matrix

Setup and Notation. Let $S = \{v_i\}_{i=1}^N \subset \mathbb{R}^d$ be a set of input vectors. Consider a parameter matrix $W \in \mathbb{R}^{d \times d}$ and a loss function $\mathcal{L}(W; v_i)$. We define the per-sample gradient $G_i \in \mathbb{R}^{d \times d}$ as:

$$G_i := \nabla_W \mathcal{L}(W; v_i)$$